

Project Report: Student Record Management System

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1. Executive Summary

This report provides a comprehensive analysis of the 'Student Record Management System', a Python-based console application developed to manage student records efficiently. The system stores information in CSV format and provides features for adding, viewing, searching, updating, deleting, sorting, and counting student records.

2. Project Overview

The Student Record Management System is a lightweight application developed in Python. It demonstrates file handling, modular programming, menu-driven interfaces, and data management using CSV files.

2.1 Objectives

1. Store student records digitally.
2. Add, search, update and delete records.
3. Display all student information.
4. Sort records alphabetically.
5. Maintain data permanently using CSV files.

2.2 Technology Stack

- Language: Python 3.x
- Database: CSV File (students.csv)
- Libraries: csv, os
- IDE: Visual Studio Code / PyCharm / IDLE

3. Architecture and Design

The application follows a modular structure where each operation is implemented through separate Python functions. The CSV file acts as the data storage layer, while the main menu controls user interaction.

3.1 Functional Modules

- Add Student
- View Students
- Search Student
- Update Student
- Delete Student
- Sort Students
- Total Students

4. Technical Analysis

The system uses Python file handling and CSV operations to read and write student records. Conditional statements, loops, and functions ensure efficient record management. The modular design makes the project easy to maintain and extend.

5. Conclusion and Future Improvements

The Student Record Management System successfully demonstrates CRUD operations using Python. Future improvements include a Tkinter GUI, SQLite/MySQL database integration, login authentication, attendance management, fee management, and report generation.